

Diagnosis information - Front-end electronics

Power supply KESSY control unit

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Power or ground supply faulty

Fault setting condition(s)

- Supply voltage < 9 V
- Supply voltage > 15 V

Fault effects

KESSY control unit No function

Diagnosis information - Front-end electronics

Antenna Short circuit to B+/ground or open circuit in the line

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line

Fault effects

- Vehicle access system (KESSY): Function not available or available only to a limited extent

Diagnosis information - Front-end electronics

Proximity sensor Short circuit to B+/ground or open circuit in the line

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line

Fault effects

Vehicle access system (KESSY) via **Proximity sensor** No function

Diagnosis information - Front-end electronics

Proximity sensor Short circuit to B+/ground or open circuit in the line

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line

Fault effects

Vehicle access system (KESSY) via **Proximity sensor** No function

Diagnosis information - Front-end electronics

Proximity sensor Short circuit to B+/ground or open circuit in the line

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line

Fault effects

Vehicle access system (KESSY) via **Proximity sensor** No function

Diagnosis information - Front-end electronics

Proximity sensor Short circuit to B+/ground or open circuit in the line

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line

Fault effects

Vehicle access system (KESSY) via **Proximity sensor** No function

Diagnosis information - Front-end electronics

Door handle too often actuated

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- **Door handle** > 30 actuated

without **Door** open

Fault effects

Door open individual Vehicle access system (KESSY) **not** possible

Diagnosis information - Front-end electronics

Button Door exterior permanent actuated

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- **Door exterior** The button is stuck or permanently actuated. > 30 Second(s)

Fault effects

Door close individual with **Button not** possible

Diagnosis information - Front-end electronics

Button Door exterior permanent actuated

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- **Door exterior** The button is stuck or permanently actuated. > 30 Second(s)

Fault effects

Door close individual with **Button not** possible

Diagnosis information - Front-end electronics

Button Door exterior permanent actuated

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- **Door exterior** The button is stuck or permanently actuated. > 30 Second(s)

Fault effects

Door close individual with **Button not** possible

Diagnosis information - Front-end electronics

Button Door exterior permanent actuated

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- **Door exterior** The button is stuck or permanently actuated. > 30 Second(s)

Fault effects

Door close individual with **Button not** possible

Diagnosis information - Front-end electronics

Ignition key **Battery empty**

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Vehicle open with **Remote control**

Fault effects

Vehicle open / close with **Remote control**
not possible

Diagnosis information - Front-end electronics

Ignition key not synchronized

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Vehicle open with **Remote control**
- "Open" button on key pressed too often outside range actuated

(see operating instructions "Emergency key")

Fault effects

Vehicle open with **Remote control**
not possible

Diagnosis information - Front-end electronics

Floor light(s) Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Floor light(s) Faulty

Fault effects

Floor light(s) No function

Diagnosis information - Front-end electronics

Lamp Front lid Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

Lamp Front lid No function

C11000

Diagnosis information - Front-end electronics

Power supply KESSY control unit

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Power or ground supply faulty

Fault setting condition(s)

- Supply voltage < 9 V
- Supply voltage > 15 V

Fault effects

KESSY control unit No function

C11001

Diagnosis information - Front-end electronics

Power supply KESSY control unit

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Power or ground supply faulty

Fault setting condition(s)

- Supply voltage < 9 V
- Supply voltage > 15 V

Fault effects

KESSY control unit No function

C11005

Diagnosis information - Front-end electronics

Front-end electronics Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

- **Front-end electronics** No function

C11006

Diagnosis information - Control module Front-end electronics

Front-end electronics, Coding

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- No/incorrect coding

Fault effects

- ♦ **Control module Front-end electronics** Limited function

Diagnosis information - Front-end electronics

Electrically adjustable steering column (EVLS) no/incorrect coding/calibration

▲ **CAUTION**

Adjustment of the **Electrically adjustable steering column (EVLS) !**
Intervention during the adjustment can cause injuries to the hands or fingers.

♦ Switch the ignition off or disconnect the power supply for the **Electrically adjustable steering column (EVLS)** .

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ **Electrically adjustable steering column (EVLS)** not coded
- ♦ **Electrically adjustable steering column (EVLS)** not calibrated

Fault effects

Electrically adjustable steering column (EVLS) No function

Diagnosis information - Front-end electronics

Wiper no communication

⚠ CAUTION

Wiper motor can come on!

Danger of injury to hands and fingers when working on the wiper mechanism when the engine is running!

- ◆ Switch ignition off and/or disconnect the power supply for the wiper motor.

NOTE

Risk of damage to parts!

Risk of damage to paintwork on the hood and wiper arms.

- ◆ The wiper arms can touch the hood if the hood is open and the wiper motor is running.
- ◆ Do not open the hood/engine compartment lid when the wiper arms are raised - risk of damage to paintwork and mechanism.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ◆ No power supply for the **Wiper** or only limited power supply available.
- ◆ Open circuit in LIN line between the **Front-end electronics** control unit and the **Wiper**
- ◆ **Wiper** faulty

Fault setting condition(s)

- No communication with the **Wiper** for $t > 4 \text{ s}$

Fault effects

- **Wiper**: Function not available or available only to a limited extent

Diagnosis information - Front-end electronics

Rain sensor no communication

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Rain sensor** coded

Possible fault causes

- ♦ No power supply for the **Rain sensor** or only limited power supply available.
- ♦ Open circuit in LIN line between the **Front-end electronics** control unit and the **Rain sensor**
- ♦ **Rain sensor** faulty

Fault setting condition(s)

- No communication with the **Rain sensor** for $t > 4$ s

Fault effects

- **Rain sensor** No function

Diagnosis information - Front-end electronics

Wiper no communication

⚠ CAUTION

Wiper motor can come on!

Danger of injury to hands and fingers when working on the wiper mechanism when the engine is running!

- ◆ Switch ignition off and/or disconnect the power supply for the wiper motor.

NOTE

Risk of damage to parts!

Risk of damage to paintwork on the hood and wiper arms.

- ◆ The wiper arms can touch the hood if the hood is open and the wiper motor is running.
- ◆ Do not open the hood/engine compartment lid when the wiper arms are raised - risk of damage to paintwork and mechanism.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ◆ No power supply for the **Wiper** or only limited power supply available.
- ◆ Open circuit in LIN line between the **Front-end electronics** control unit and the **Wiper**
- ◆ **Wiper** faulty

Fault setting condition(s)

- No communication with the **Wiper** for $t > 4 \text{ s}$

Fault effects

- **Wiper**: Function not available or available only to a limited extent

Diagnosis information - Front-end electronics

Sensor faulty

Diagnostic conditions

- The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

- Sensor does not function

Diagnosis information - Front-end electronics

Wiper no communication

⚠ CAUTION

Wiper motor can come on!

Danger of injury to hands and fingers when working on the wiper mechanism when the engine is running!

- ◆ Switch ignition off and/or disconnect the power supply for the wiper motor.

NOTE

Risk of damage to parts!

Risk of damage to paintwork on the hood and wiper arms.

- ◆ The wiper arms can touch the hood if the hood is open and the wiper motor is running.
- ◆ Do not open the hood/engine compartment lid when the wiper arms are raised - risk of damage to paintwork and mechanism.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ◆ No power supply for the **Wiper** or only limited power supply available.
- ◆ Open circuit in LIN line between the **Front-end electronics** control unit and the **Wiper**
- ◆ **Wiper** faulty

Fault setting condition(s)

- No communication with the **Wiper** for $t > 4$ s

Fault effects

- **Wiper**: Function not available or available only to a limited extent

Diagnosis information - Front-end electronics

Rain sensor no communication

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Rain sensor** coded

Possible fault causes

- ♦ No power supply for the **Rain sensor** or only limited power supply available.
- ♦ Open circuit in LIN line between the **Front-end electronics** control unit and the **Rain sensor**
- ♦ **Rain sensor** faulty

Fault setting condition(s)

- No communication with the **Rain sensor** for $t > 4 \text{ s}$

Fault effects

- **Rain sensor** No function

Diagnosis information - Front-end electronics

Wiper no communication

⚠ CAUTION

Wiper motor can come on!

Danger of injury to hands and fingers when working on the wiper mechanism when the engine is running!

- ◆ Switch ignition off and/or disconnect the power supply for the wiper motor.

NOTE

Risk of damage to parts!

Risk of damage to paintwork on the hood and wiper arms.

- ◆ The wiper arms can touch the hood if the hood is open and the wiper motor is running.
- ◆ Do not open the hood/engine compartment lid when the wiper arms are raised - risk of damage to paintwork and mechanism.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ◆ No power supply for the **Wiper** or only limited power supply available.
- ◆ Open circuit in LIN line between the **Front-end electronics** control unit and the **Wiper**
- ◆ **Wiper** faulty

Fault setting condition(s)

- No communication with the **Wiper** for $t > 4 \text{ s}$

Fault effects

- **Wiper**: Function not available or available only to a limited extent

Diagnosis information - Front-end electronics

Rain sensor no communication

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Rain sensor** coded

Possible fault causes

- ♦ No power supply for the **Rain sensor** or only limited power supply available.
- ♦ Open circuit in LIN line between the **Front-end electronics** control unit and the **Rain sensor**
- ♦ **Rain sensor** faulty

Fault setting condition(s)

- No communication with the **Rain sensor** for $t > 4$ s

Fault effects

- **Rain sensor** No function

C13401

Diagnosis information - Front-end electronics

Wiper no communication

⚠ CAUTION

Wiper motor can come on!

Danger of injury to hands and fingers when working on the wiper mechanism when the engine is running!

- ◆ Switch ignition off and/or disconnect the power supply for the wiper motor.

NOTE

Risk of damage to parts!

Risk of damage to paintwork on the hood and wiper arms.

- ◆ The wiper arms can touch the hood if the hood is open and the wiper motor is running.
- ◆ Do not open the hood/engine compartment lid when the wiper arms are raised - risk of damage to paintwork and mechanism.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ◆ No power supply for the **Wiper** or only limited power supply available.
- ◆ Open circuit in LIN line between the **Front-end electronics** control unit and the **Wiper**
- ◆ **Wiper** faulty

Fault setting condition(s)

- No communication with the **Wiper** for $t > 4 \text{ s}$

Fault effects

- **Wiper**: Function not available or available only to a limited extent

Diagnosis information - Front-end electronics

Rain sensor no communication

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Rain sensor** coded

Possible fault causes

- ♦ No power supply for the **Rain sensor** or only limited power supply available.
- ♦ Open circuit in LIN line between the **Front-end electronics** control unit and the **Rain sensor**
- ♦ **Rain sensor** faulty

Fault setting condition(s)

- No communication with the **Rain sensor** for $t > 4$ s

Fault effects

- **Rain sensor** No function